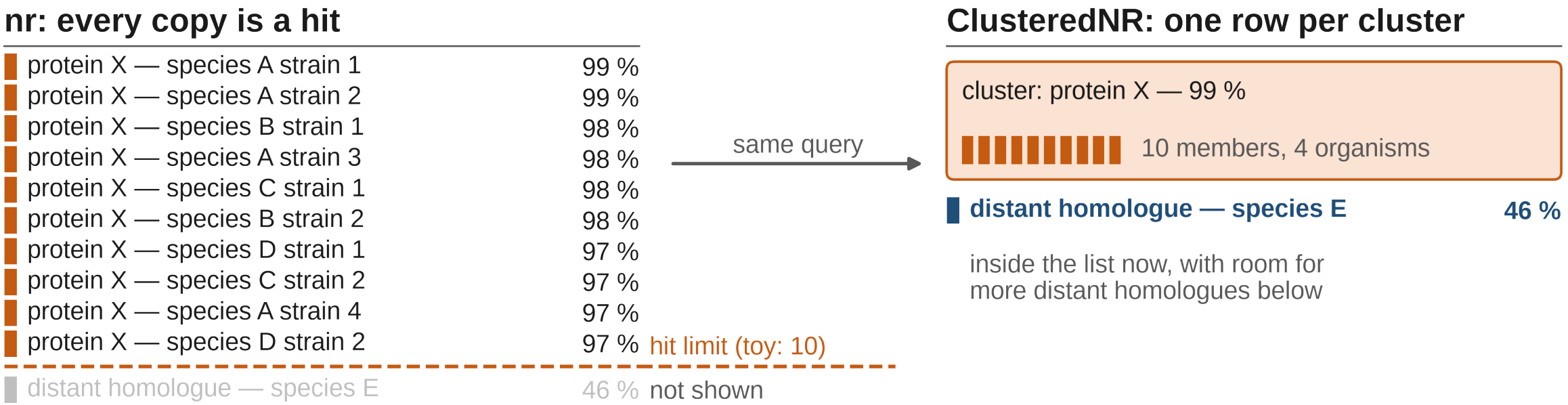
NCBI BLAST in 2026

Which database?

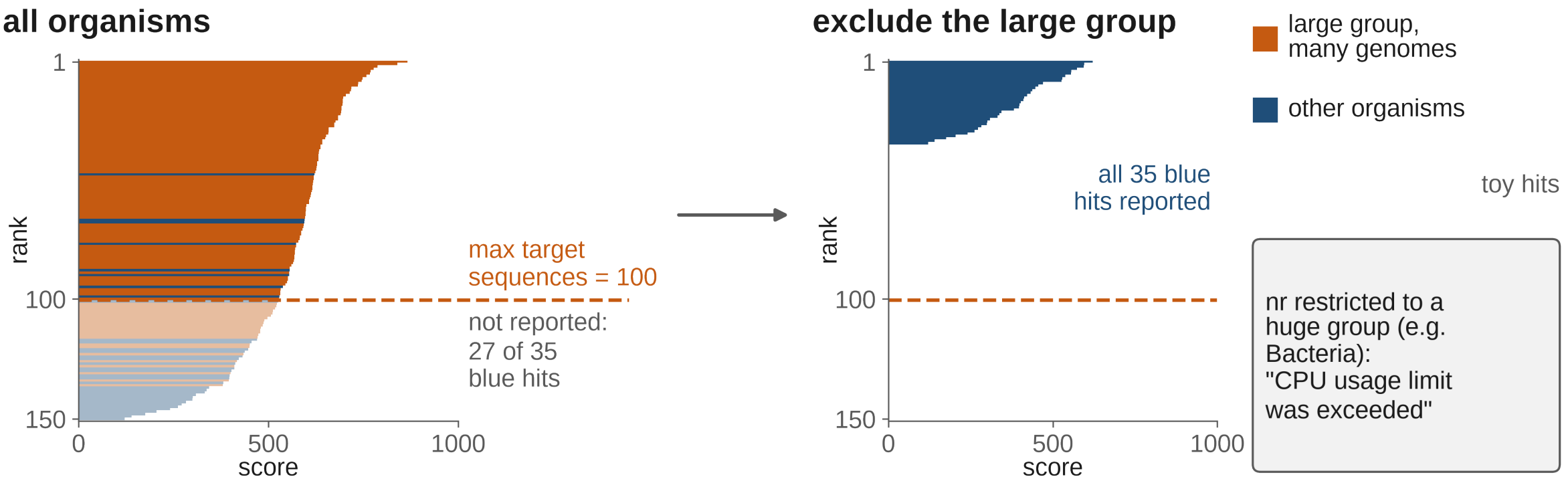
- **core_nt** (default for nucleotides): GenBank without most eukaryotic chromosomes
- **ClusteredNR** (default for proteins): one representative per cluster of proteins >90 % identical



toy hits

What the hit list does not show

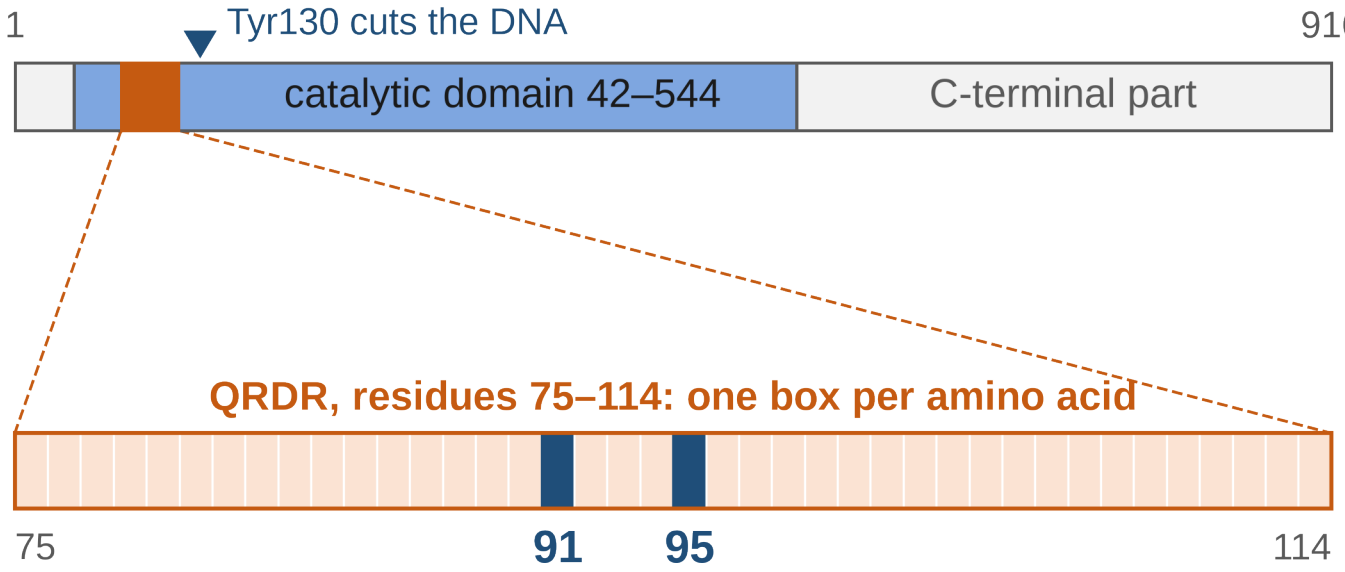
- "Max target sequences" cuts the list: 100 hits by default
- Organism filter: restrict to a group, or **exclude** a group
- The form remembers the last filter: click **Reset page** first

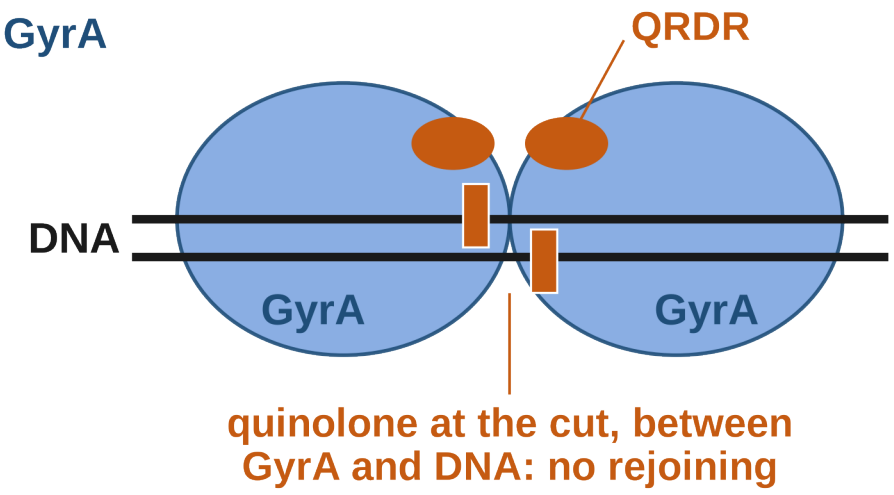


Exercises 1.7–1.8 – Antibiotic resistance

Quinolones block DNA gyrase

- DNA gyrase relaxes supercoils ahead of the replication fork
- Ciprofloxacin traps gyrase on cut DNA; the cell dies
- Resistance mutations cluster in the **QRDR** of GyrA

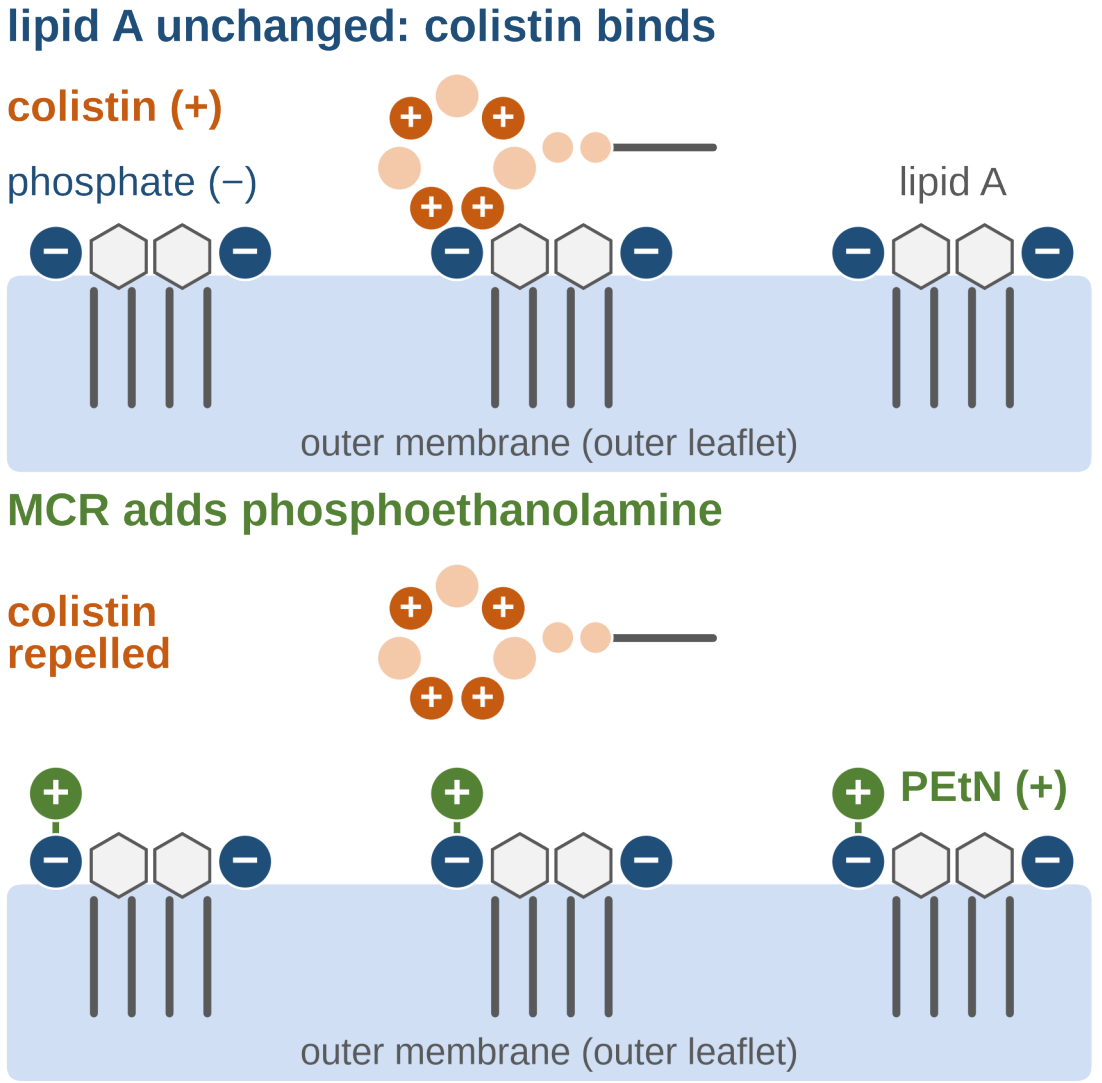




UniProt Q9JYX5; QRDR: Yoshida et al. 1990 Antimicrob Agents Chemother 34:1271; drug site: Aldred et al. 2014 Biochemistry 53:1565

Colistin attacks the outer membrane

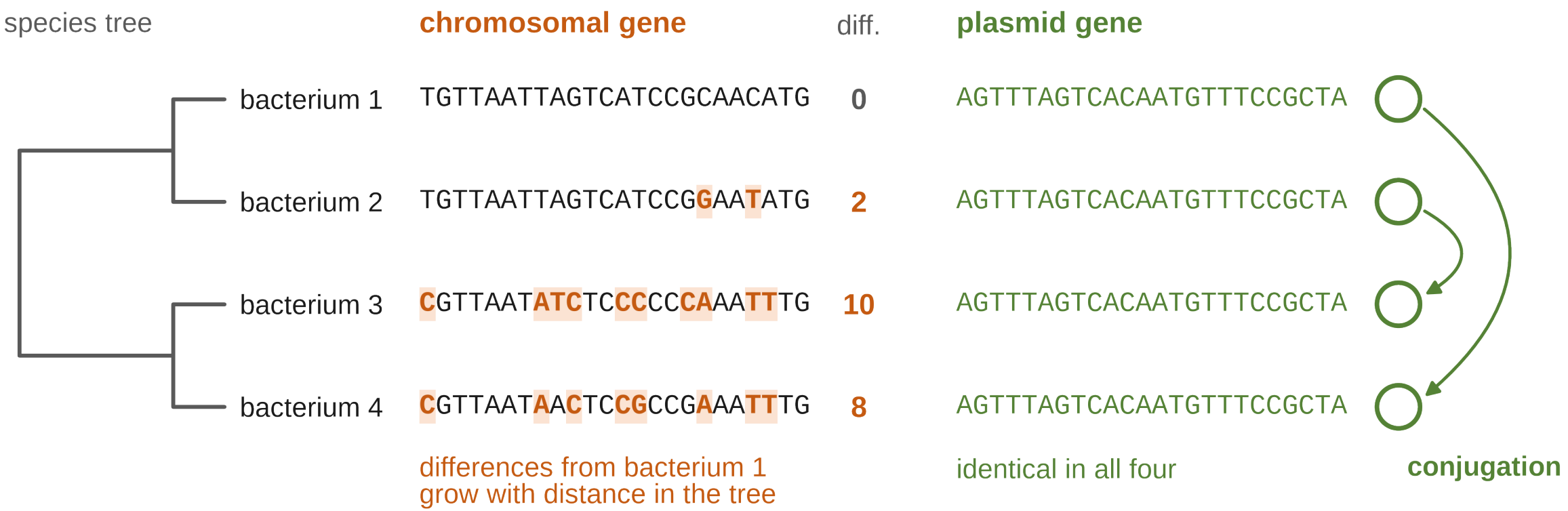
- Colistin (polymyxin E): positively charged peptide
- It binds the negative phosphates of **lipid A**
- MCR adds **phosphoethanolamine** (+) to lipid A
- Less negative charge → colistin binds less



Olaitan et al. 2014 Front Microbiol 5:643; Liu et al. 2016 Lancet Infect Dis 16:161

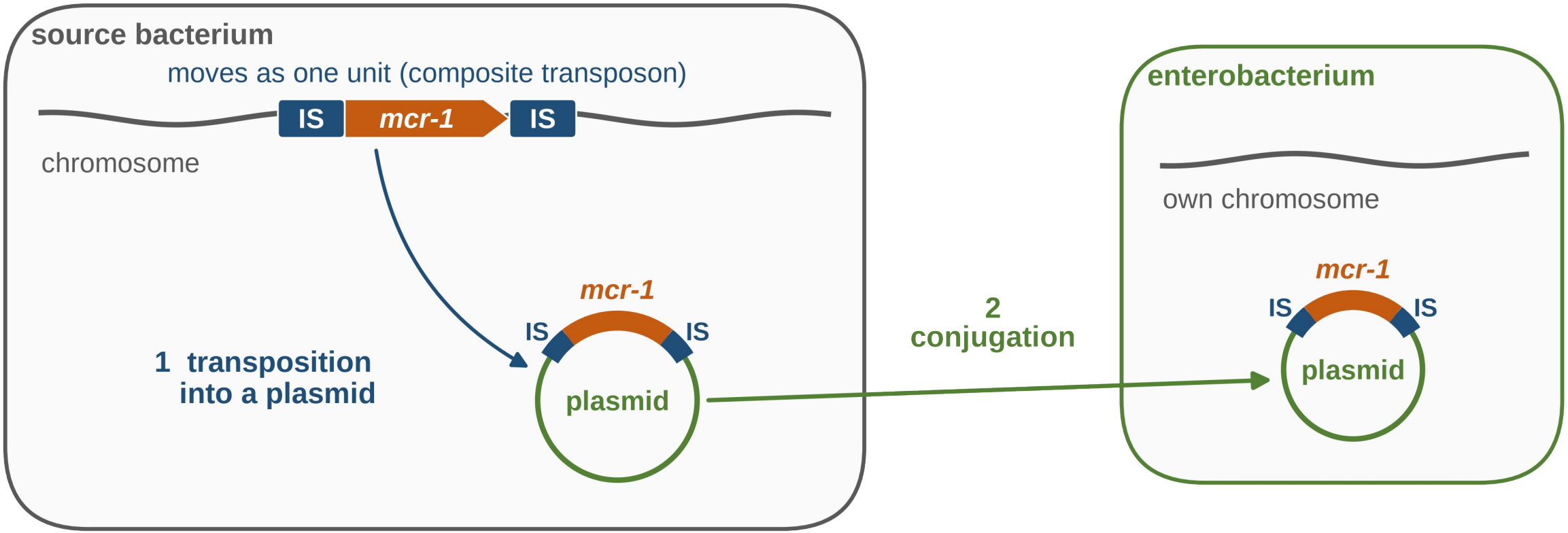
Genes move between bacteria

- Chromosomal genes are inherited from mother to daughter cell
- Plasmids can transfer between unrelated bacteria (**conjugation**)



Mobile elements pick up chromosomal genes

- An insertion sequence (IS) on each side of a gene can move it as one unit
- Chromosome of the source bacterium → plasmid → other species

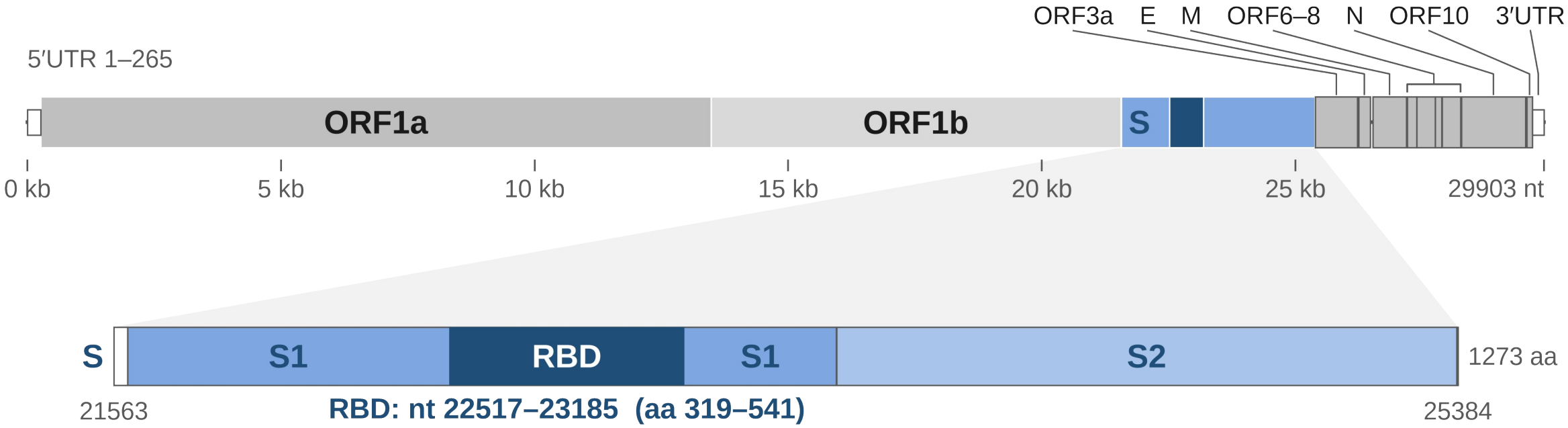


Snesrud et al. 2016 Antimicrob Agents Chemother, doi:10.1128/AAC.01457-16

Exercise 1.9 – SARS-CoV-2

The SARS-CoV-2 genome

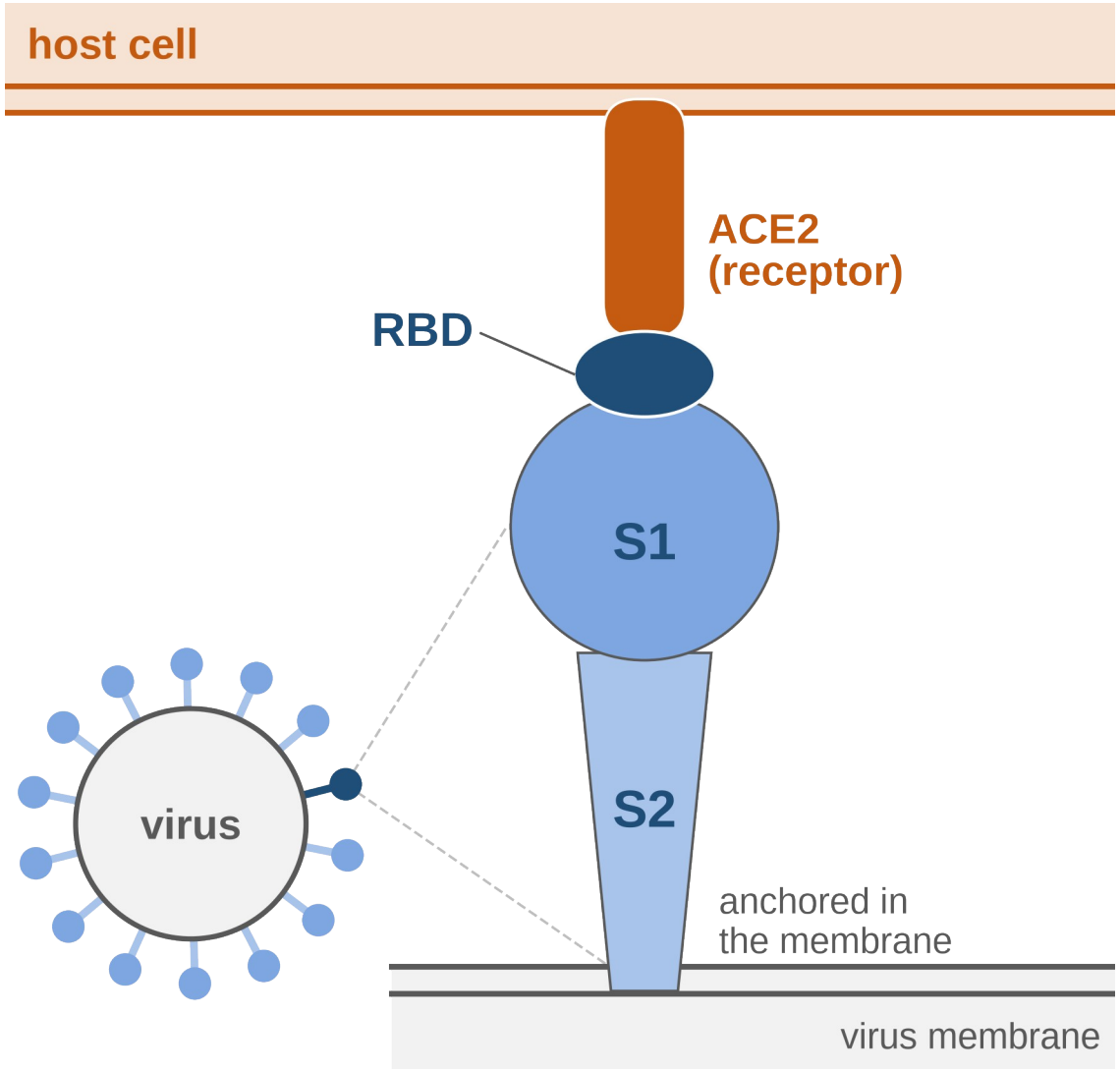
- 30 kb RNA genome; ORF1ab encodes the replication machinery
- The spike protein (S) binds the host cell receptor



NCBI RefSeq NC_045512.2 annotation; S1/S2/RBD from UniProt P0DTC2

The RBD decides which cells the virus enters

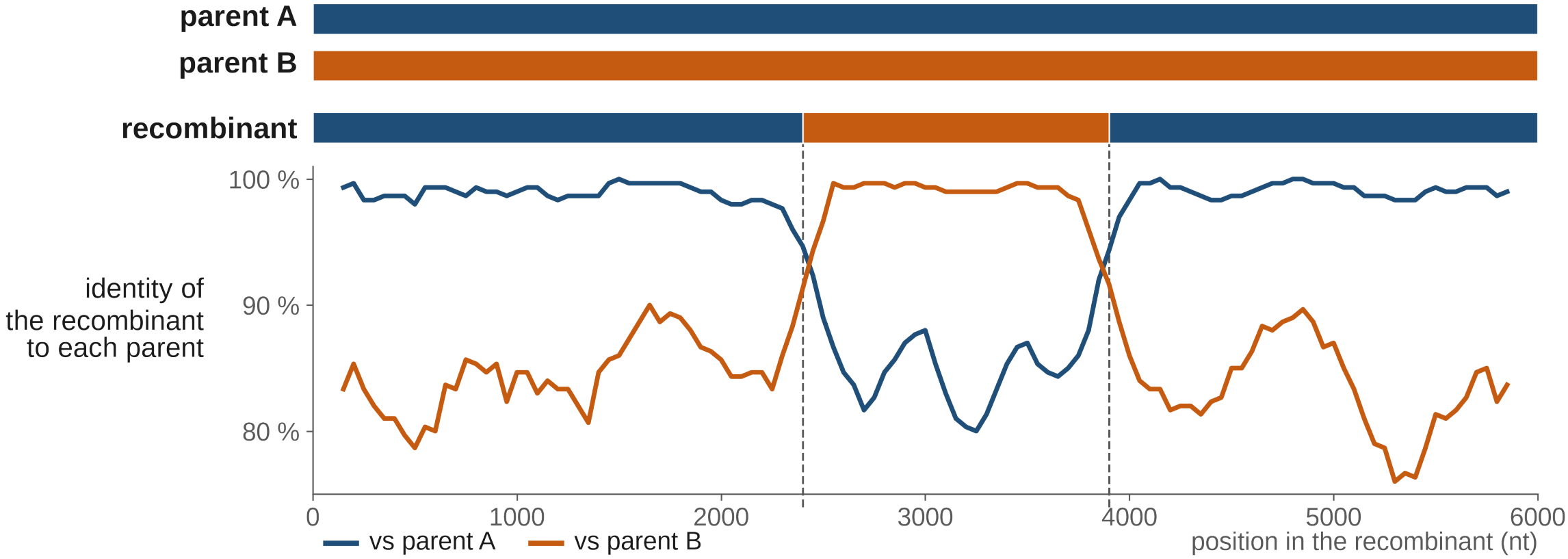
- The receptor-binding domain (RBD) of S binds **ACE2**
- A few RBD residues decide binding to human ACE2
- Changes here matter more than elsewhere in the genome



domain boundaries from UniProt P0DTC2

Recombination makes mosaic genomes

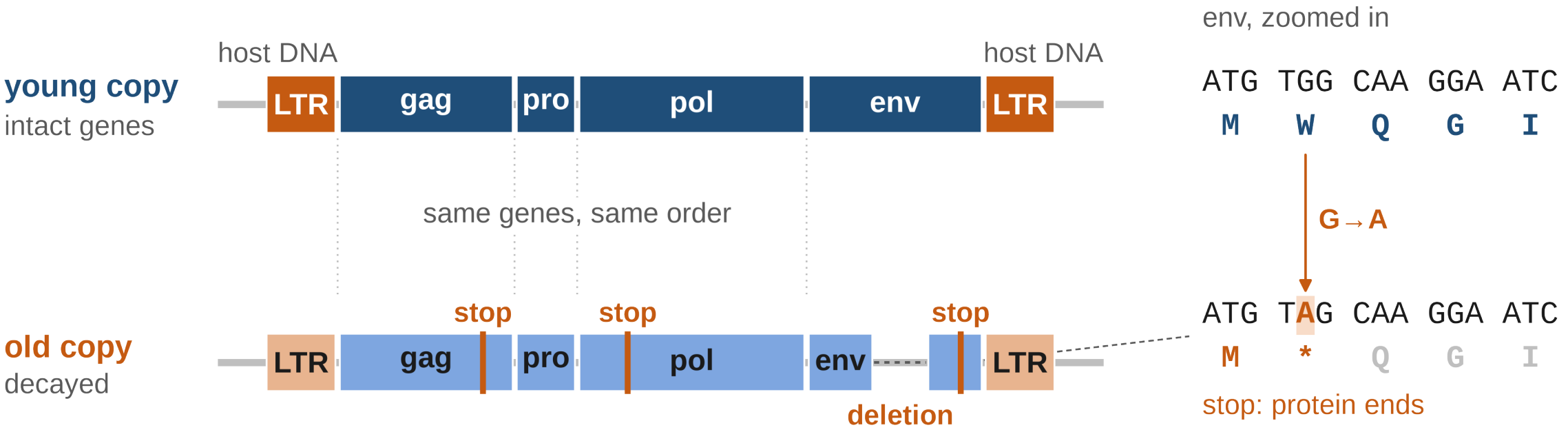
- Two coronaviruses infecting the same animal can swap genome parts
- The closest relative of a recombinant depends on the region



Exercise 1.10 – Syncytin

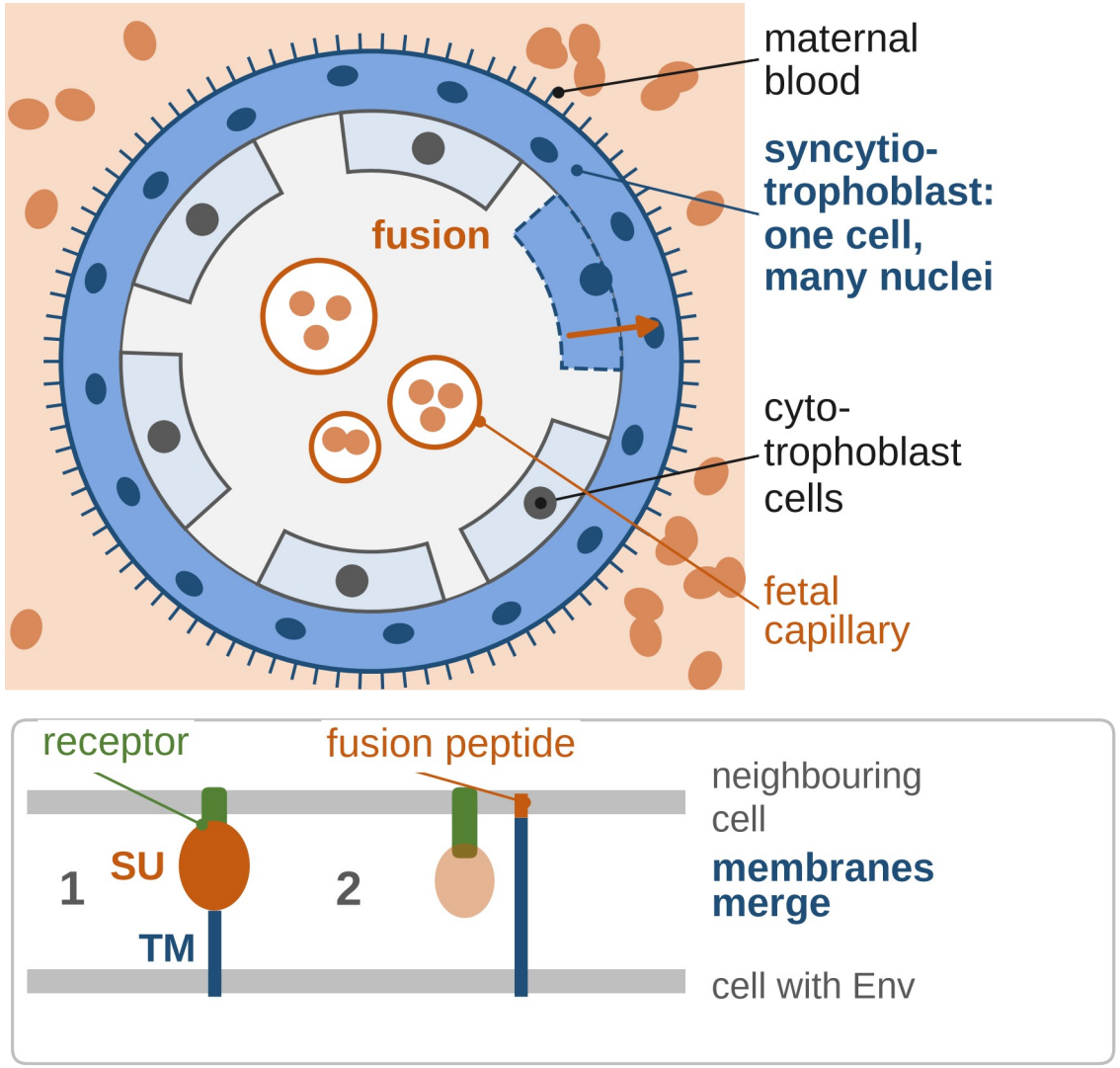
Retroviruses that became part of our genome

- A retrovirus copies its RNA into DNA and inserts it into the host chromosome
- Inserted into a germ cell, the provirus is inherited (**endogenous retrovirus**)
- ~8 % of the human genome; most copies are broken by mutations



Syncytiotrophoblast: cells fused into one

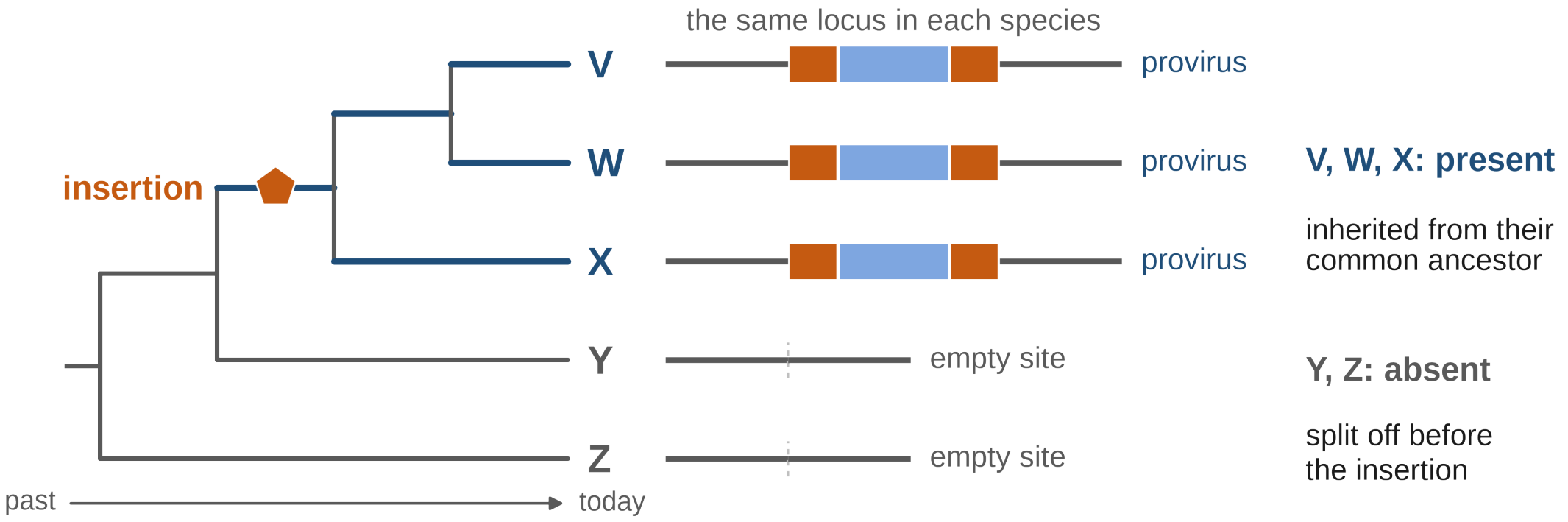
- The placenta's outer layer is one huge multinucleate cell
- It forms by **fusion** of trophoblast cells
- Retroviral Env: SU binds the receptor, TM fuses the membranes



Lokossou et al. 2014, Viruses 6:4609; UniProt Q9UQF0

Dating an insertion with a species tree

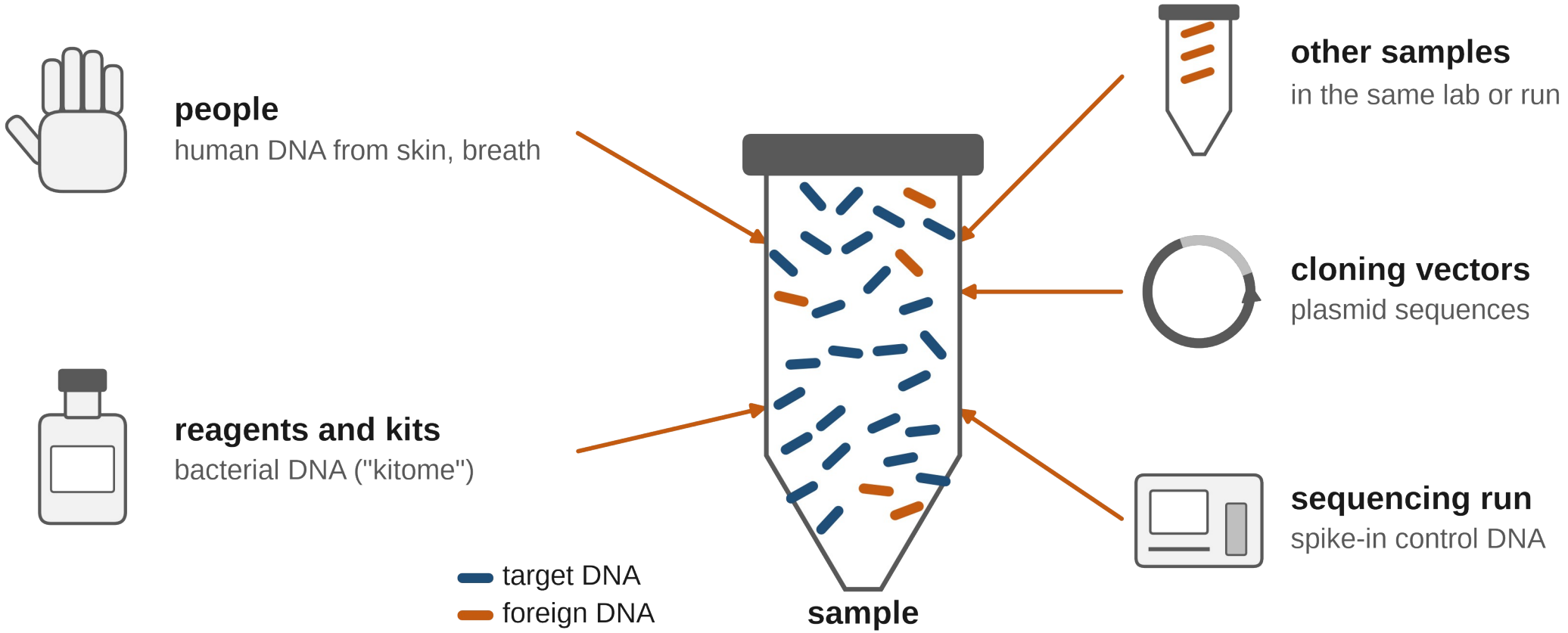
- An insertion present in all descendants of an ancestor happened before they split
- Species that split off earlier have an empty site



Exercise 1.11 – Contamination

Where foreign DNA comes from

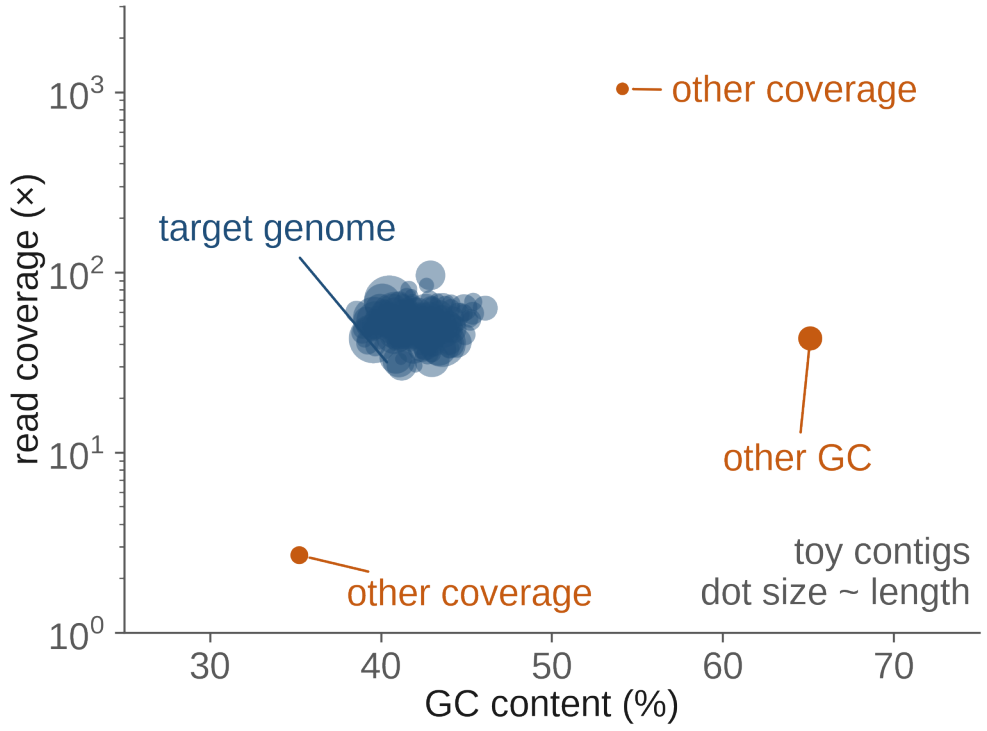
- People, reagents, other samples, vectors, sequencing controls



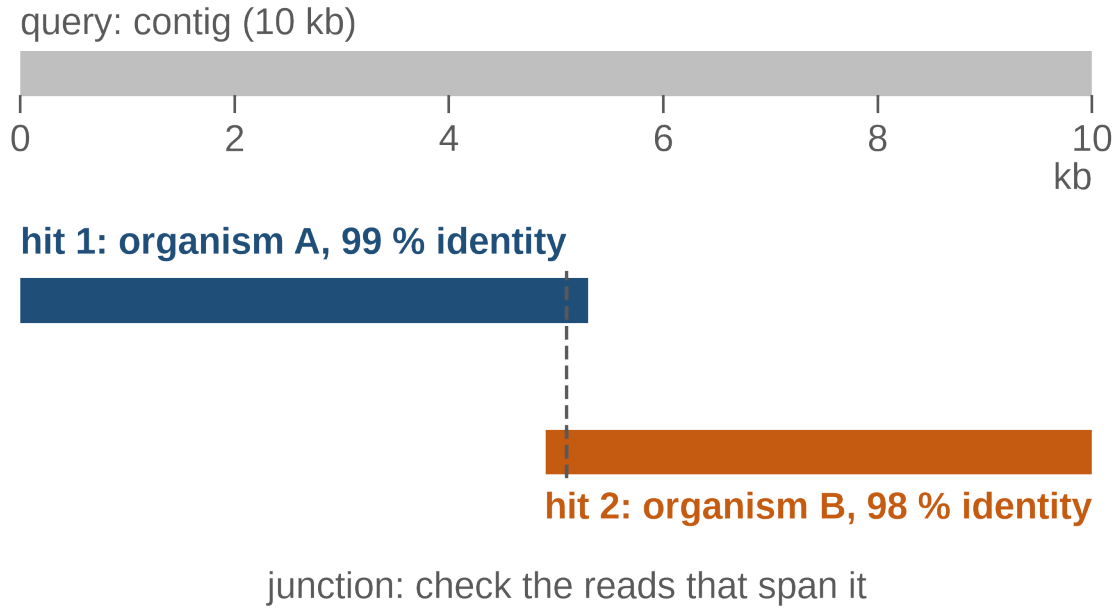
Signs of a contaminated contig

- Best hit from another organism, coverage unlike the rest of the genome
- A contig matching two organisms in different parts: possible **chimera**

one genome = one cloud



one contig, two organisms



toy example